



Yakkshit

AI & Robotics Engineer | Full-Stack & Autonomous Systems Specialist

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Summary

Versatile AI/Robotics Engineer and Full-Stack Developer with a solid foundation in autonomous systems, multi-agent swarm robotics, and scalable cloud architectures. Experienced in designing real-time perception/control pipelines in C++ and Python (ROS/ROS2), deploying production RAG and AI agent systems with LangChain, and fine-tuning open-source LLMs (Qwen, Llama). Proven leadership in bridging cutting-edge robotics hardware with modern web and AI applications.

Experience

- Research Assistant – Swarm Robotics & Multi-Agent Systems

2026-02 - Present

University of Konstanz

Researching robotic clusters, swarm intelligence, and emergent decentralized multi-agent behaviors under a global Research Excellence initiative.
- Autonomous Navigation & Perception Engineer

2025-08 - Present

BRT (Bodensee Racing Team - Formula Student)

Developing autonomous perception, obstacle avoidance, and path-planning systems for the Formula Student driverless racecar.
- Software Developer Intern – AI & Full-Stack

2025-01 - 2025-07

Nutrish.ai

Engineered an AI-driven personal nutrition agent leveraging predictive modeling and Retrieval-Augmented Generation.
- Lead Full-Stack Developer

2023-12 - 2024-12

Circleup AG

Led core system architecture, microservices migration, and machine learning model integration for circular economy platforms.
- Founder & AI Engineer

2022-06 - Present

Cedzlabs

Building developer productivity platforms and custom LLM fine-tuning pipelines.

Projects

- Custom AI Resume Generator & Cloud API (Qwen2.5 / Unsloth)

Python

FastAPI

Unsloth

LoRA

GGUF

Docker

Hugging Face

Fine-tuned Qwen2.5 models using LoRA/Unsloth, exported to 4-bit GGUF, and deployed an OpenAI-compatible FastAPI backend with streaming PDF rendering.
- Full-Stack Agentic Coder (Multi-Agent System)

Python

LangChain

Multi-Agent Systems

Tool-Calling

LLMs

Engineered a multi-agent AI system with specialized planning, coding, and debugging agents utilizing advanced tool-calling and prompt workflows.
- Android Multilingual Translator App

Java

Android SDK

Google Translate API

Material UI

Developed a native Android mobile application for real-time speech and text translation integrating Google Translate API and offline cache.

ARCore vs WebXR Spatial Computing Benchmarking

Unity ARCore WebXR Spatial Computing C#

Benchmarked CPU load, rendering latency, and memory footprint across ARCore and WebXR spatial web frameworks for spatial computing applications.

Education

Universität Konstanz

2025-04 - Present

Master of Science in Computer and Information Science in AI for Robotics, Human-Robot Collaboration & Multi-Agent Systems

GPA: Current Studies

JNTU Kakinada & Blekinge Institute of Technology (BTH)

2020-08 - 2024-06

Bachelor of Science in Computer Science in Software Engineering & Spatial Computing (Thesis: ARCore vs WebXR)

GPA: First Class with Distinction

Achievements

Deep Learning Specialization

DeepLearning.AI / Coursera – Neural Networks, CNNs & Sequence Models

ROS2 & Advanced Autonomous Robotics

Comprehensive Navigation Stack, URDF & Sensor Fusion Architecture

Languages

English (Fluent / Professional - C1)

German / Deutsch (Grundkenntnisse - A2)

Telugu (Native)

Skills

Python

C++

ROS / ROS2

Gazebo & Webots

Multi-Agent Systems

LLM Fine-Tuning (LoRA/Unsloth)

RAG & LangChain

PyTorch

Computer Vision (OpenCV)

Sensor Fusion (LiDAR/IMU)

Next.js & React

TypeScript & JavaScript

Java & Android SDK

AWS & Docker

PostgreSQL & Vector DBs

Embedded C++ & 3D Prototyping

English (Fluent - C1)

German (A2 / Grundkenntnisse)

Telugu (Native)